COMP1002 Final Assignment

Biogas Feedstock Collection Network: Data Structures and Algorithms for Sustainable Waste Management

Semester 2, 2026 @ EECMS, Curtin University

1. Introduction

A renewable energy company is investigating opportunities to convert organic waste and by-products from the red meat industry into renewable energy through anaerobic digestion and biogas production.

Potential feedstock sources, such as farms, are geographically distributed across Australia. Efficiently managing information about these farms and planning potential feedstock collection activities requires appropriate data structures and algorithms.

In this assignment, you will design and implement a backend **Biogas Feedstock Collection Network** that manages farms information, represents transportation connections, prioritises feedstock collection requests, and produces collection performance reports.

The system consists of four integrated modules:

Module	Name	Role in Biogas Feedstock Network
1	Graph-Based Farm Network	Models the transportation network between farms and the proposed biogas hub
2	Hash-Based Farm Lookup	Stores and retrieves farms information efficiently
3	Heap-Based Feedstock Collection Scheduling	Prioritises feedstock collection requests using feedstock availability and transportation efficiency
4	Sorting Collection Records	Sorts collection records for end-of-period reporting

The purpose of the assignment is not to assess your knowledge of biogas engineering, agricultural production, or geographic information systems. The real-world scenario provides the context in which you apply data structures and algorithms.

2. Learning Objectives

This assignment aims to improve students' understanding and application of fundamental data structures and algorithms in solving a real-world sustainability problem.

By completing this assignment, you should be able to:

- Implement and analyse an adjacency-list graph with BFS, DFS, and Dijkstra's shortest-path algorithm.
- Design and implement a hash table with collision handling suitable for at least 40 records.
- Implement a scheduler that combines feedstock availability and estimated transportation time.
- Implement Merge Sort and Quick Sort and benchmark their performance under different input conditions.
- Integrate multiple data structures and algorithms into a coherent, testable system.
- Analyse the time and space complexity of implemented algorithms.
- Justify appropriate data structure and algorithm choices for a real-world computational problem.

3. Assignment Scenario and Dataset

3.1 Real-World Context

The assignment is inspired by a real-world challenge involving the identification and collection of geographically distributed organic feedstock sources for potential use in a biogas system.

The following public farm-location website provides the geographic context for the assignment:

Farm Transparency Project — Map

The website provides publicly available geographic information about farm locations across Australia. You are not assessed on web scraping, geographic information systems, or data collection techniques.

3.2 Collecting Farm Data

You must use the farm data from the above public farm-location website. Requirements:

- **One region.** Choose a single state/territory, or one contiguous area within a state (e.g., the South West of WA), and collect the farms data from that region. Your choice of region does not affect the marking criteria.